

Kinco MK043E-20DT

2-Axis Stepper Motor Positioning

MODBUS RTU RS-485 Interface

Hardware: NEMA23 stepper motors + MD-2545 drivers

Master: ESP32 (MODBUS RTU Master)

Protocol: MODBUS RTU, 8N1, Station ID 1

Software: KincoBuilder IL (Instruction List)

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1. Hardware Overview

The Kinco MK043E-20DT PLC controls two NEMA23 stepper motors via MD-2545 drivers. Positioning instructions (PHOME, PABS, PSTOP, PREL, PJOG) execute on the PLC using its internal PTO (Pulse Train Output) channels.

Communication with the master controller (ESP32) uses MODBUS RTU over RS-485 (COM1). The ESP32 writes command and parameter values to the PLC's %V memory area; the PLC ladder logic executes the corresponding motion functions.

PTO Channel Assignment

PTO Channel	Axis ID	Step/Pulse	Direction	Pos. Register
PTO0	0	%Q0.0	%Q0.2	%SMD212
PTO1	1	%Q0.1	%Q0.3	%SMD242

Pulse Frequency Range

125 Hz to 200 kHz per channel (per KincoBuilder help).

2. Physical I/O Map

Digital Outputs (8 total: %Q0.0 - %Q0.7)

Output	Symbol	Function	Status
%Q0.0	A0_STEP	Step/Pulse - Axis 0	Used by PTO0 (internal)
%Q0.1	A1_STEP	Step/Pulse - Axis 1	Used by PTO1 (internal)
%Q0.2	A0_DIR	Direction - Axis 0	Used by PTO0 (internal)
%Q0.3	A1_DIR	Direction - Axis 1	Used by PTO1 (internal)
%Q0.4	A0_ENABLE_OUT	Driver Enable - Axis 0	Controlled by %V100.0
%Q0.5	A1_ENABLE_OUT	Driver Enable - Axis 1	Controlled by %V200.0
%Q0.6	(free)	-	Available
%Q0.7	(free)	-	Available

Digital Inputs (used)

Input	Symbol	Function
%I0.0	A0_HOME_SENSOR	HOME sensor - Axis 0
%I0.1	A1_HOME_SENSOR	HOME sensor - Axis 1

Note: %Q0.0 through %Q0.3 must NOT be used as regular latch coils. They are internally driven by PHOME/PABS/PREL/PJOE blocks. %Q0.6 and %Q0.7 are free for additional outputs.

3. MODBUS Register Map - Summary

The ESP32 communicates as MODBUS RTU Master. The PLC is configured as MODBUS RTU Slave (Station ID 1). All motion parameters and commands are mapped to holding registers (4xxxx). Addresses assume an addressing base of 0 (40001 = address 0).

Axis 0 - MODBUS Addresses

MODBUS	PLC	Type	Description
40051	%VW100	WORD	Control word
40053-40054	%VD104	DINT	PABS target position
40055-40056	%VD108	DWORD	PABS max frequency
40057	%VW112	WORD	PABS min frequency
40058	%VW114	WORD	PABS accel time (ms)
40059	%VW116	INT	HOME mode
40060	%VW118	INT	HOME direction
40061	%VW120	WORD	HOME min frequency
40063-40064	%VD124	DWORD	HOME max frequency
40065	%VW128	WORD	HOME accel time (ms)
40071-40072	%VD140	DINT	PREL relative distance
40073-40074	%VD144	DWORD	PREL max frequency
40075	%VW148	WORD	PREL min frequency
40076	%VW150	WORD	PREL accel time (ms)
40077-40078	%VD152	DWORD	PJOG speed
40079	%VW156	INT	PJOG direction (internal)
40081	%VW160	WORD	Status word 1
40082	%VW162	WORD	Status word 2 (reserved)
40083-40084	%VD164	DINT	Current position

Axis 1 - MODBUS Addresses

MODBUS	PLC	Type	Description
40101	%VW200	WORD	Control word
40103-40104	%VD204	DINT	PABS target position
40105-40106	%VD208	DWORD	PABS max frequency
40107	%VW212	WORD	PABS min frequency
40108	%VW214	WORD	PABS accel time (ms)
40109	%VW216	INT	HOME mode
40110	%VW218	INT	HOME direction
40111	%VW220	WORD	HOME min frequency
40113-40114	%VD224	DWORD	HOME max frequency
40115	%VW228	WORD	HOME accel time (ms)
40121-40122	%VD240	DINT	PREL relative distance
40123-40124	%VD244	DWORD	PREL max frequency
40125	%VW248	WORD	PREL min frequency
40126	%VW250	WORD	PREL accel time (ms)
40127-40128	%VD252	DWORD	PJOG speed

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MODBUS	PLC	Type	Description
40129	%VW256	INT	PJOG direction (internal)
40131	%VW260	WORD	Status word 1
40132	%VW262	WORD	Status word 2 (reserved)
40133-40134	%VD264	DINT	Current position

4. Control Word - Axis 0

Register: %VW100 / MODBUS 40051

Control Bits

Bit	PLC Name	Value with Enable	Function
0	%V100.0 / Ctrl_Enable	0x0001	Enable driver (must be ON for any motion)
1	%V100.1 / Ctrl_ResetPos	0x0003	Reset logical position (clear %SMD212)
2	%V100.2 / Ctrl_StartPABS	0x0005	Start absolute positioning (edge-triggered)
3	%V100.3 / Ctrl_StartHome	0x0009	Start homing (edge-triggered)
4	%V100.4 / Ctrl_ResetStatus	0x0011	Reset internal status and errors
5	%V100.5 / Ctrl_Stop	0x0021	Stop axis (calls PSTOP)
6	%V100.6 / Ctrl_StartPREL	0x0041	Start relative positioning (edge-triggered)
7	%V100.7 / Ctrl_JogForward	0x0081	Jog forward (level-triggered)
8	%V101.0 / Ctrl_JogBackward	0x0101	Jog backward (level-triggered)

Important Notes

- PABS, HOME, and PREL are edge-triggered: ESP32 must set the bit HIGH, wait ~100 ms, then set it back LOW.
- Jog Forward and Jog Backward are level-triggered: keep bit HIGH while moving, set LOW to stop.
- Never set JogForward and JogBackward simultaneously (logic ignores both if so).
- After any edge-triggered command, reset word to 0x0001 (Enable only).
- Default enable state after startup: write 0x0001 to the control word.

5. Control Word - Axis 1

Register: %VW200 / MODBUS 40101

Control Bits

Bit	PLC Name	Value with Enable	Function
0	%V200.0 / Ctrl_Enable	0x0001	Enable driver (must be ON for any motion)
1	%V200.1 / Ctrl_ResetPos	0x0003	Reset logical position (clear %SMD242)
2	%V200.2 / Ctrl_StartPABS	0x0005	Start absolute positioning (edge-triggered)
3	%V200.3 / Ctrl_StartHome	0x0009	Start homing (edge-triggered)
4	%V200.4 / Ctrl_ResetStatus	0x0011	Reset internal status and errors
5	%V200.5 / Ctrl_Stop	0x0021	Stop axis (calls PSTOP)
6	%V200.6 / Ctrl_StartPREL	0x0041	Start relative positioning (edge-triggered)
7	%V200.7 / Ctrl_JogForward	0x0081	Jog forward (level-triggered)
8	%V201.0 / Ctrl_JogBackward	0x0101	Jog backward (level-triggered)

Important Notes

- PABS, HOME, and PREL are edge-triggered: ESP32 must set the bit HIGH, wait ~100 ms, then set it back LOW.
- Jog Forward and Jog Backward are level-triggered: keep bit HIGH while moving, set LOW to stop.
- Never set JogForward and JogBackward simultaneously (logic ignores both if so).
- After any edge-triggered command, reset word to 0x0001 (Enable only).
- Default enable state after startup: write 0x0001 to the control word.

6. Parameter Registers - Axis 0

6.1 PABS Parameters

MODBUS	PLC	Type	Description	Default
40053-40054	%VD104	DINT	Target absolute position (pulses)	0
40055-40056	%VD108	DWORD	Max frequency (Hz)	2000
40057	%VW112	WORD	Min frequency (Hz)	300
40058	%VW114	WORD	Accel/Decel time (ms)	300

6.2 HOME Parameters

MODBUS	PLC	Type	Description	Default
40059	%VW116	INT	Mode (0=HOME+NHOME, 1=HOME only)	1
40060	%VW118	INT	Direction (0=Forward, 1=Backward)	0
40061	%VW120	WORD	Min frequency (Hz)	200
40063-40064	%VD124	DWORD	Max frequency (Hz)	1000
40065	%VW128	WORD	Accel/Decel time (ms)	300

6.3 PREL Parameters

MODBUS	PLC	Type	Description	Default
40071-40072	%VD140	DINT	Relative distance (pulses, signed)	0
40073-40074	%VD144	DWORD	Max frequency (Hz)	2000
40075	%VW148	WORD	Min frequency (Hz)	300
40076	%VW150	WORD	Accel/Decel time (ms)	300

6.4 PJOG Parameters

MODBUS	PLC	Type	Description	Default
40077-40078	%VD152	DWORD	Jog speed / frequency (Hz)	500
40079	%VW156	INT	Direction (0=Forward, 1=Backward, auto-set)	0

7. Parameter Registers - Axis 1

7.1 PABS Parameters

MODBUS	PLC	Type	Description	Default
40103-40104	%VD204	DINT	Target absolute position (pulses)	0
40105-40106	%VD208	DWORD	Max frequency (Hz)	2000
40107	%VW212	WORD	Min frequency (Hz)	300
40108	%VW214	WORD	Accel/Decel time (ms)	300

7.2 HOME Parameters

MODBUS	PLC	Type	Description	Default
40109	%VW216	INT	Mode (0=HOME+NHOME, 1=HOME only)	1
40110	%VW218	INT	Direction (0=Forward, 1=Backward)	0
40111	%VW220	WORD	Min frequency (Hz)	200
40113-40114	%VD224	DWORD	Max frequency (Hz)	1000
40115	%VW228	WORD	Accel/Decel time (ms)	300

7.3 PREL Parameters

MODBUS	PLC	Type	Description	Default
40121-40122	%VD240	DINT	Relative distance (pulses, signed)	0
40123-40124	%VD244	DWORD	Max frequency (Hz)	2000
40125	%VW248	WORD	Min frequency (Hz)	300
40126	%VW250	WORD	Accel/Decel time (ms)	300

7.4 PJOG Parameters

MODBUS	PLC	Type	Description	Default
40127-40128	%VD252	DWORD	Jog speed (Hz)	500
40129	%VW256	INT	Direction (0=Forward, 1=Backward, auto-set)	0

8. Status Word - Axis 0

Register: %VW160 / MODBUS 40081 (read-only by ESP32)

Bit	PLC	Name	Description
0	%V160.0	HomeOK	Homing completed successfully
1	%V160.1	HomeDone	PHOME DONE flag
2	%V160.2	HomeErr	PHOME error flag
3	%V160.3	PabsDone	PABS completed
4	%V160.4	PabsErr	PABS error
5	%V160.5	AxisBusy	Axis busy (homing OR PABS OR PREL OR PJOG active)
6	%V160.6	HomingActive	Homing in progress
7	%V160.7	PabsActive	PABS in progress
8	%V161.0	HomeSensor	Physical HOME sensor state (%I0.0)
9	%V161.1	SystemReady	System ready for movement commands
10	%V161.2	EnableOut	Driver enable output state (%Q0.4)
11	%V161.3	StopDone	PSTOP completed
12	%V161.4	PrelDone	PREL completed
13	%V161.5	PrelErr	PREL error
14	%V161.6	JogActive	PJOG in progress
15	%V161.7	JogErr	PJOG error

AxisBusy combines: HomingActive OR PabsActive OR PrelActive OR JogActive.

SystemReady requires: Enable ON + HomeOK + no HOME error + no PABS error + no PREL error + no PJOG error.

9. Status Word - Axis 1

Register: %VW260 / MODBUS 40131 (read-only by ESP32)

Bit	PLC	Name	Description
0	%V260.0	HomeOK	Homing completed successfully
1	%V260.1	HomeDone	PHOME DONE flag
2	%V260.2	HomeErr	PHOME error flag
3	%V260.3	PabsDone	PABS completed
4	%V260.4	PabsErr	PABS error
5	%V260.5	AxisBusy	Axis busy (homing OR PABS OR PREL OR PJOG active)
6	%V260.6	HomingActive	Homing in progress
7	%V260.7	PabsActive	PABS in progress
8	%V261.0	HomeSensor	Physical HOME sensor state (%I0.1)
9	%V261.1	SystemReady	System ready for movement commands
10	%V261.2	EnableOut	Driver enable output state (%Q0.5)
11	%V261.3	StopDone	PSTOP completed
12	%V261.4	PrelDone	PREL completed
13	%V261.5	PrelErr	PREL error
14	%V261.6	JogActive	PJOG in progress
15	%V261.7	JogErr	PJOG error

AxisBusy combines: HomingActive OR PabsActive OR PrelActive OR JogActive.

SystemReady requires: Enable ON + HomeOK + no HOME error + no PABS error + no PREL error + no PJOG error.

10. Status Word 2 - Both Axes

Reserved registers, declared but no bits assigned in current program.

Axis	PLC	MODBUS
0	%VW162	40082
1	%VW262	40132

11. Current Position Registers

Axis	PLC	MODBUS	Type	Source
0	%VD164	40083-40084	DINT	Copy of %SMD212 (PTO0 position)
1	%VD264	40133-40134	DINT	Copy of %SMD242 (PTO1 position)

Updated every scan cycle (%SM0.0 always ON). Units: pulses.

12. Error ID Registers

Axis	PLC	Function
0	%VB140	PABS error code
0	%VB141	HOME error code
0	%VB142	STOP error code
0	%VB170	PREL error code
0	%VB171	JOG error code
1	%VB240	PABS error code

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Axis	PLC	Function
1	%VB241	HOME error code
1	%VB242	STOP error code
1	%VB270	PREL error code
1	%VB271	JOG error code

Error codes follow KincoBuilder positioning instruction spec. Refer to KincoBuilder help for specific code meanings per instruction type.

13. MODBUS Command Sequences - Axis 0

13.1 HOME Sequence (Axis 0)

Step	Action	MODBUS Write / Read
1	Enable driver	40051 = 0x0001
2	Set HOME mode	40059 = 1
3	Set HOME direction	40060 = 0
4	Set HOME min freq	40061 = 200
5	Set HOME max freq	40063-40064 = 1000
6	Set HOME accel time	40065 = 300
7	Trigger HOME (edge UP)	40051 = 0x0009
8	Trigger HOME (edge DOWN)	40051 = 0x0001
9	Wait for completion	Read 40081 until bit 0=1 or bit 2=1

13.2 PABS Sequence (Axis 0)

Step	Action	MODBUS Write / Read
1	Set target position	40053-40054 = target (DINT, pulses)
2	Set max frequency	40055-40056 = MAXF (DWORD, Hz)
3	Set min frequency	40057 = MINF (WORD, Hz)
4	Set accel time	40058 = TIME (WORD, ms)
5	Trigger PABS (edge UP)	40051 = 0x0005
6	Trigger PABS (edge DOWN)	40051 = 0x0001
7	Wait for completion	Read 40081 until bit 3=1 or bit 4=1
8	Read current position	40083-40084

13.3 PREL Sequence (Axis 0)

Step	Action	MODBUS Write / Read
1	Set relative distance	40071-40072 = distance (DINT, signed)
2	Set max frequency	40073-40074 = MAXF (DWORD, Hz)
3	Set min frequency	40075 = MINF (WORD, Hz)
4	Set accel time	40076 = TIME (WORD, ms)
5	Trigger PREL (edge UP)	40051 = 0x0041
6	Trigger PREL (edge DOWN)	40051 = 0x0001
7	Wait for completion	Read 40081 until bit 12=1 or bit 13=1

13.4 PJOG Sequence (Axis 0)

Step	Action	MODBUS Write / Read
1	Set jog speed	40077-40078 = speed (DWORD, Hz)
2	Start forward jog	40051 = 0x0081 (hold)
3	Stop jog	40051 = 0x0001
4	Start backward jog	40051 = 0x0101 (hold)
5	Emergency stop	40051 = 0x0021

13.5 Command Values Quick Reference (Axis 0)

Command	40051 Value
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Command	40051 Value
Enable only	0x0001
Reset position	0x0003
Start PABS	0x0005
Start HOME	0x0009
Reset status	0x0011
Stop (PSTOP)	0x0021
Start PREL	0x0041
Jog forward	0x0081
Jog backward	0x0101

14. MODBUS Command Sequences - Axis 1

14.1 HOME Sequence (Axis 1)

Step	Action	MODBUS Write / Read
1	Enable driver	40101 = 0x0001
2	Set HOME mode	40109 = 1
3	Set HOME direction	40110 = 0
4	Set HOME min freq	40111 = 200
5	Set HOME max freq	40113-40114 = 1000
6	Set HOME accel time	40115 = 300
7	Trigger HOME (edge UP)	40101 = 0x0009
8	Trigger HOME (edge DOWN)	40101 = 0x0001
9	Wait for completion	Read 40131 until bit 0=1 or bit 2=1

14.2 PABS Sequence (Axis 1)

Step	Action	MODBUS Write / Read
1	Set target position	40103-40104 = target (DINT, pulses)
2	Set max frequency	40105-40106 = MAXF (DWORD, Hz)
3	Set min frequency	40107 = MINF (WORD, Hz)
4	Set accel time	40108 = TIME (WORD, ms)
5	Trigger PABS (edge UP)	40101 = 0x0005
6	Trigger PABS (edge DOWN)	40101 = 0x0001
7	Wait for completion	Read 40131 until bit 3=1 or bit 4=1
8	Read current position	40133-40134

14.3 PREL Sequence (Axis 1)

Step	Action	MODBUS Write / Read
1	Set relative distance	40121-40122 = distance (DINT, signed)
2	Set max frequency	40123-40124 = MAXF (DWORD, Hz)
3	Set min frequency	40125 = MINF (WORD, Hz)
4	Set accel time	40126 = TIME (WORD, ms)
5	Trigger PREL (edge UP)	40101 = 0x0041
6	Trigger PREL (edge DOWN)	40101 = 0x0001
7	Wait for completion	Read 40131 until bit 12=1 or bit 13=1

14.4 PJOG Sequence (Axis 1)

Step	Action	MODBUS Write / Read
1	Set jog speed	40127-40128 = speed (DWORD, Hz)
2	Start forward jog	40101 = 0x0081 (hold)
3	Stop jog	40101 = 0x0001
4	Start backward jog	40101 = 0x0101 (hold)
5	Emergency stop	40101 = 0x0021

14.5 Command Values Quick Reference (Axis 1)

Command	40101 Value
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Command	40101 Value
Enable only	0x0001
Reset position	0x0003
Start PABS	0x0005
Start HOME	0x0009
Reset status	0x0011
Stop (PSTOP)	0x0021
Start PREL	0x0041
Jog forward	0x0081
Jog backward	0x0101

15. Internal Data Types (IL)

KincoBuilder IL uses prefixed constants to match the target register type. Type mismatches cause compile errors.

Type	Signed	Range	IL Prefix	Example
INT	Yes	-32768 to 32767	I#	MOVE I#1, %VW116
WORD	No	0 to 65535	W#	MOVE W#300, %VW108
DINT	Yes	-2 ³¹ to 2 ³¹ -1	DI#	MOVE DI#0, %VD100
DWORD	No	0 to 2 ³² -1	DW#	MOVE DW#2000, %VD104

Note: INT and WORD both occupy 16 bits but differ in signed interpretation. DINT and DWORD both occupy 32 bits but differ in signed interpretation. HOME MODE and DIRC expect INT. PABS POS expects DINT. MAXF expects DWORD. MINF and TIME expect WORD.

16. Wiring Diagram - Kinco MK043E-20DT to MD-2545

Axis 0 (Motor 1)

Kinco Output	Signal	MD-2545 Driver Input
%Q0.0	STEP / PUL	PUL+ / PUL-
%Q0.2	DIR	DIR+ / DIR-
%Q0.4	ENABLE	ENA+ / ENA-
0V (common)	GND	Common / negative side

Axis 1 (Motor 2)

Kinco Output	Signal	MD-2545 Driver Input
%Q0.1	STEP / PUL	PUL+ / PUL-
%Q0.3	DIR	DIR+ / DIR-
%Q0.5	ENABLE	ENA+ / ENA-
0V (common)	GND	Common / negative side

Home Sensors

Kinco Input	Sensor	Type
%I0.0	HOME Axis 0	NPN or PNP (configurable)
%I0.1	HOME Axis 1	NPN or PNP (configurable)

MODBUS RS-485 (COM1)

Parameter	Value
Protocol	MODBUS RTU Slave
Station ID	1
Baudrate	9600 or 19200
Data format	8N1 (8 data, no parity, 1 stop)
Physical	RS-485 2-wire on COM1

IMPORTANT: Verify endianness when using 32-bit registers (DINT/DWORD). Some MODBUS libraries or HMIs swap high/low words. Test with a known value before production use.